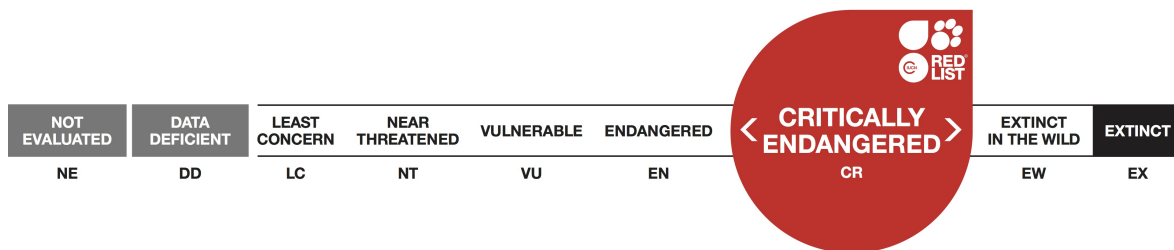


## *Hogna ingens*, Desertas Wolf Spider

Assessment by: Cardoso, P.



View on [www.iucnredlist.org](http://www.iucnredlist.org)

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## Taxonomy

| Kingdom  | Phylum     | Class     | Order   | Family    |
|----------|------------|-----------|---------|-----------|
| Animalia | Arthropoda | Arachnida | Araneae | Lycosidae |

**Taxon Name:** *Hogna ingens* (Blackwall, 1857)

**Synonym(s):**

- *Lycosa ingens* Blackwall, 1857

**Common Name(s):**

- English: Desertas Wolf Spider

**Taxonomic Source(s):**

Platnick, N.I. 2014. The World Spider Catalog, Version 14.5. P. Merrett & H.D. Cameron (eds). American Museum of Natural History. Available at: <http://research.amnh.org/iz/spiders/catalog/index.html>. (Accessed: 31 March 2014).

**Taxonomic Notes:**

*Hogna ingens* was described in 1857 by John Blackwall (1857). Despite several taxonomic redescrptions from pioneer arachnologists (Blackwall 1867, Kulczynski 1899, Roewer 1960, Denis 1962 and later Wunderlich 1992), every other aspect of this remarkable species remained unknown until recently (Crespo *et al.* 2014).

## Assessment Information

**Red List Category & Criteria:** Critically Endangered B1ab(ii,iii,v)+2ab(ii,iii,v) [ver 3.1](#)

**Year Published:** 2014

**Date Assessed:** April 7, 2014

**Justification:**

The Desertas Islands (Madeira, Portugal) are the sole home of one of the largest and rarest wolf spider species worldwide, *Hogna ingens* (Blackwall, 1857) (Araneae, Lycosidae). Despite its size, it inhabits a single valley in the north of the Deserta Grande Island, Vale da Castanheira, currently invaded by the non-native grass *Phalaris aquatica*. This invasive species competes with the native flora and was subject to several eradication experiments, namely through fire and chemicals. The current distribution of *H. ingens* covers 23 ha, a recent reduction from its original 83 ha, corresponding to the entire valley. A total of 4,447 and 4,086 adults were estimated to live in the valley during 2011 and 2012, respectively. We found a significant negative impact of *P. aquatica* cover on the presence and abundance of *H. ingens* and that chemical treatment specifically directed towards the plant may be the only way to effectively recover the spider's habitat. We suggest as future measures: (1) regular monitoring of the spider's population; (2) possibly extend chemical treatments to other areas (subject to tests on side-effects to other native species); and (3) *ex situ* conservation with future reintroduction of adult specimens. Based upon the small geographic range of the species with only one location and continuing decline of its habitat and area of occupancy, it is assessed as Critically Endangered.

## Geographic Range

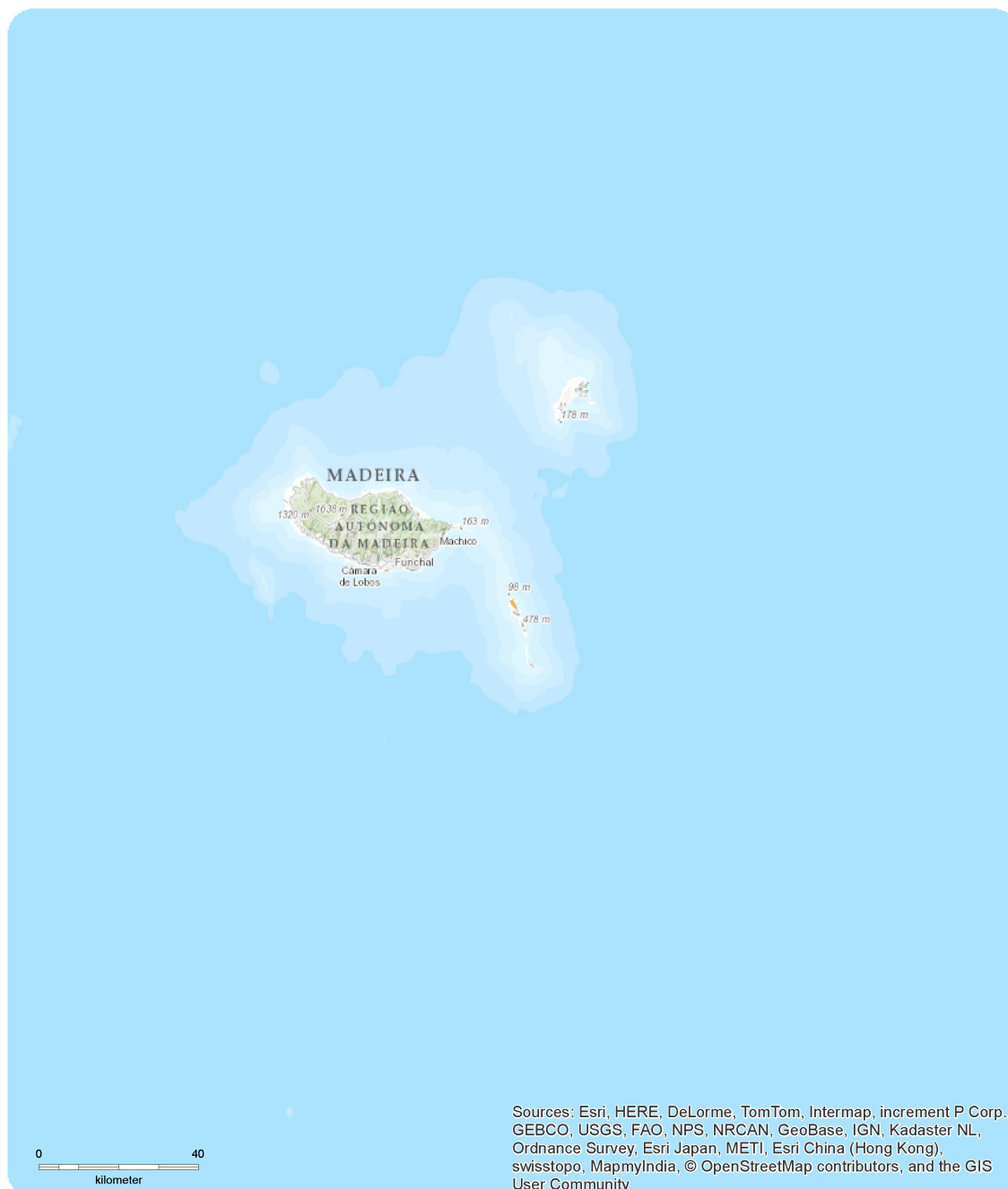
### Range Description:

*Hogna ingens* is known to historically occur only in a valley at the northern end of Deserta Grande Island (Desertas, Madeira, Portugal). The Vale da Castanheira is approximately 2.8 km long and its width varies between 180 m to 400 m. The estimated area is 83 ha. The elevation of the valley ranges from 150 up to 350 m asl.

### Country Occurrence:

**Native:** Portugal (Madeira)

## Distribution Map

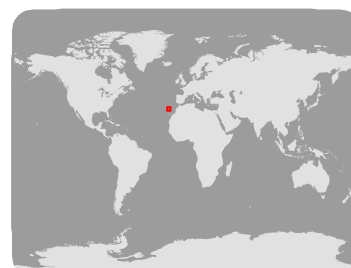


### *Hogna ingens*

Range

■ Extant (resident)

Compiled by:  
IUCN



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.



## Population

The estimated population size of *H. ingens* was 4,447 adult specimens in 2011 and 4,086 in 2012.

**Current Population Trend:** Decreasing

## Habitat and Ecology (see Appendix for additional information)

The Vale da Castanheira, single site of occurrence of the species, was originally occupied by a mix of native grasses and bare rock/soil. Most of it is now occupied by *Phalaris aquatica*, an invasive grass that recently expanded (expansion detected in 2005). This invasive plant covers the entire surface of the soil and crevices, preventing the accessibility to shelters that are usually occupied by the spider as well as other endemic fauna.

In the absence of native terrestrial mammals, this spider with a 40 mm body length is a top predator in its small habitat, and although its major prey consists of other invertebrates, such as the staphylinid beetle *Ocypus olens* (Müller, 1764) or the invasive millipede *Ommatoiulus moreleti* (Lucas, 1860), adults have even been seen predating on juveniles of the lizard *Lacerta dugesii mauli* Mertens, 1938. The latter, along with birds and mice, should nevertheless be the major predators of *H. ingens*, mostly during its juvenile stage. It is in this stage that the spider is most vulnerable to predators, because adding to its smaller size, it tends to disperse in order to find new shelters, thus maximizing the likelihood of encounters with potential predators (including conspecific adults). As spiders grow and find proper shelters (mostly below rocks but also in soil crevices), their dispersal propensity gradually decreases. It takes about two years for a spider to reach maturity.

**Systems:** Terrestrial

## Use and Trade (see Appendix for additional information)

The species is not known to be in the pet trade, but individuals are collected from the site to photograph and/or keep in private collections.

## Threats (see Appendix for additional information)

The Vale da Castanheira is currently mostly covered by a non-native invasive grass, *Phalaris aquatica* L.. The settlement of this grass was hidden for some years, due to the presence of rabbits (*Oryctolagus cuniculus* L.), who grazed and stalled the proliferation of the grass. Given that rabbits were eradicated from the Vale da Castanheira in 1996, *P. aquatica* lost its main consumer and now proliferates. This grass appears to not only displace the native herbs, but also many of the native animals, as one of the most common arthropods is now the invader diplopod *Ommatoiulus moreleti* (Lucas, 1860). Also, *P. aquatica* covers the surface of the soil and rocks, making the microhabitats below the rocks harder to access for arthropods, which take shelter there during daytime.

## Conservation Actions (see Appendix for additional information)

We suggest as future conservation measures:

(1) Regular monitoring of the trends of the distribution and population size of the species. This will provide more robust data for ecological and conservation studies. Although this is already done on a voluntary basis, it is not regular or exhaustive.

(2) Taking into account possible side-effects on other species (an aspect which needs to be studied further), to extend the preliminary chemical treatments to the areas currently affected by the presence of *P. aquatica*. This could prove to be essential to remove the invasive grass from Vale da Castanheira, and may prove to also be beneficial to other endemic and native invertebrates.

(3) Implement *ex situ* conservation measures for the species for future reintroduction of adults into the Vale da Castanheira. This could be easily and inexpensively done by the Madeira Natural Park authorities with eventual collaboration from zoos. This would ensure that several juvenile specimens would reach maturity and could be used to colonize previously unoccupied areas. The *ex situ* animals also constitute an insurance against the extinction of the species in its natural habitat. This measure would further be a way to showcase a remarkable endemic species to the visitors of Deserta Grande, thus connecting the public with a nearly unknown value of the Madeira natural patrimony.

## Credits

**Assessor(s):** Cardoso, P.

**Reviewer(s):** Hochkirch, A. & Gerlach, J.

## Bibliography

Blackwall, J. 1857. Description of the male of *Lycosa tarentuloides maderiana* Walck., and of three newly discovered species of the genus *Lycosa*. *Annals and Magazine of Natural History* 2(20): 282-287.

Blackwall, J. 1867. Notes on spiders, with descriptions of several species supposed to be new to arachnologists. *Annals and Magazine of Natural History* 3(20): 202-213.

Crespo, L.C., Silva, I., Borges, P.A.V. and Cardoso, P. 2014. Assessing the conservation status of the strict endemic Desertas wolf spider, *Hogna ingens* (Araneae, Lycosidae). *Journal for Nature Conservation*.

Denis, J. 1962. Les araignées de l'archipel de Madère (Mission du Professeur Vandel). *Publicações do Instituto de Zoologia Dr. Augusto Nobre* 79: 1-118.

IUCN. 2014. The IUCN Red List of Threatened Species. Version 2014.2. Available at: [www.iucnredlist.org](http://www.iucnredlist.org). (Accessed: 24 July 2014).

Kulczynski, W. 1899. Arachnoidea opera Rev. Schmitz collecta in insulis Maderianis et Selvages dictis. *Rozprawa i sprawozdanie wydział matematyczno-pzyrodniczy Academia umiejtnosi Cracov* 36: 319-461.

Roewer, C.F. 1960. Araneae Lycosaeformia II (Lycosidae) (Fortsetzung und Schluss). *Exploration du Parc National de l'Upemba. Mission G.F. de Witte* 55: 519-1040.

Wunderlich, J. 1992. Die Spinnen-Fauna der Makaronesischen Inseln. *Beiträge zur Araneologie* 1: 1-619.

## Citation

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## External Resources

For [Images and External Links to Additional Information](#), please see the Red List website.

# Appendix

## Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Habitat                                    | Season   | Suitability | Major Importance? |
|--------------------------------------------|----------|-------------|-------------------|
| 4. Grassland -> 4.4. Grassland - Temperate | Resident | Suitable    | Yes               |

## Use and Trade

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| End Use                            | Local | National | International |
|------------------------------------|-------|----------|---------------|
| Pets/display animals, horticulture | No    | Yes      | Yes           |

## Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Threat                                                                                                                                      | Timing    | Scope                                                                                                     | Severity            | Impact Score   |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------|---------------------|----------------|
| 8. Invasive & other problematic species & genes -><br>8.1. Invasive non-native/alien species -> 8.1.2.<br>Named species (Phalaris aquatica) | Ongoing   | Whole (>90%)                                                                                              | Very rapid declines | High impact: 9 |
|                                                                                                                                             | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation |                     |                |

## Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Conservation Actions in Place                             |
|-----------------------------------------------------------|
| In-Place Research, Monitoring and Planning                |
| Action Recovery plan: No                                  |
| Systematic monitoring scheme: Yes                         |
| In-Place Land/Water Protection and Management             |
| Conservation sites identified: Yes, over entire range     |
| Occur in at least one PA: Yes                             |
| Percentage of population protected by PAs (0-100): 91-100 |
| Area based regional management plan: No                   |
| Invasive species control or prevention: No                |
| In-Place Species Management                               |



|                                                            |
|------------------------------------------------------------|
| <b>Conservation Actions in Place</b>                       |
| Harvest management plan: No                                |
| Successfully reintroduced or introduced benignly: No       |
| Subject to ex-situ conservation: No                        |
| In-Place Education                                         |
| Subject to recent education and awareness programmes: No   |
| Included in international legislation: No                  |
| Subject to any international management/trade controls: No |

## Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| <b>Conservation Actions Needed</b>                                                                   |
| 2. Land/water management -> 2.2. Invasive/problematic species control                                |
| 2. Land/water management -> 2.3. Habitat & natural process restoration                               |
| 3. Species management -> 3.3. Species re-introduction -> 3.3.1. Reintroduction                       |
| 3. Species management -> 3.4. Ex-situ conservation -> 3.4.1. Captive breeding/artificial propagation |
| 4. Education & awareness -> 4.3. Awareness & communications                                          |
| 5. Law & policy -> 5.1. Legislation -> 5.1.1. International level                                    |
| 5. Law & policy -> 5.1. Legislation -> 5.1.2. National level                                         |
| 5. Law & policy -> 5.1. Legislation -> 5.1.3. Sub-national level                                     |

## Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                               |
|---------------------------------------------------------------|
| <b>Research Needed</b>                                        |
| 1. Research -> 1.2. Population size, distribution & trends    |
| 1. Research -> 1.3. Life history & ecology                    |
| 1. Research -> 1.6. Actions                                   |
| 2. Conservation Planning -> 2.1. Species Action/Recovery Plan |
| 2. Conservation Planning -> 2.2. Area-based Management Plan   |
| 3. Monitoring -> 3.1. Population trends                       |
| 3. Monitoring -> 3.4. Habitat trends                          |

## Additional Data Fields

|                                                                   |
|-------------------------------------------------------------------|
| <b>Distribution</b>                                               |
| Estimated area of occupancy (AOO) (km <sup>2</sup> ): 0.23        |
| Continuing decline in area of occupancy (AOO): Yes                |
| Extreme fluctuations in area of occupancy (AOO): No               |
| Estimated extent of occurrence (EOO) (km <sup>2</sup> ): 0.83     |
| Continuing decline in extent of occurrence (EOO): No              |
| Extreme fluctuations in extent of occurrence (EOO): No            |
| Number of Locations: 1                                            |
| Continuing decline in number of locations: No                     |
| Extreme fluctuations in the number of locations: No               |
| Lower elevation limit (m): 150                                    |
| Upper elevation limit (m): 350                                    |
| <b>Population</b>                                                 |
| Number of mature individuals: 4086                                |
| Continuing decline of mature individuals: Yes                     |
| Extreme fluctuations: Unknown                                     |
| Population severely fragmented: No                                |
| No. of subpopulations: 1                                          |
| Continuing decline in subpopulations: No                          |
| Extreme fluctuations in subpopulations: No                        |
| All individuals in one subpopulation: Yes                         |
| <b>Habitats and Ecology</b>                                       |
| Continuing decline in area, extent and/or quality of habitat: Yes |
| Generation Length (years): 2                                      |
| Movement patterns: Not a Migrant                                  |

## The IUCN Red List Partnership



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